

Retaining Value: Segment-Based Predictive Models for Customer Churn in Banking

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Abstract

Customer churn is an everyday problem for the banking sector, particularly in a rapidly evolving marketplace where there is greater customer expectations and an increase in digital competition. This study establishes a machine learning framework for predicting customer churn on a segment basis for High Net Worth (HNW) and Regular customers. The study used models based on Logistic Regression, Decision Tree, Random Forest and Gradient Boosting, focusing on interpretability, alignment to the business, and ability to be scalable. A cost-sensitive learning approach was employed to deal with the class imbalance and a churn risk scoring system was introduced that enabled model results to be converted into prioritized intervention strategies for banks to retain customers. Feature importance analysis indicated that the drivers of churn were different between segments, validating the modeling (HNW and Regular) on a segment basis. The framework developed here compromises a balance between predictive performance and explainability and is feasibly scalable to implement in practice for use in CRM systems. This study expands the literature addressing criteria for predicting customer churn in the banking context to include a replicable, interpretable, and data-driven approach to understand customer churn using a proactive approach and further, the approach taken considers personal retention strategies in the modern banking for customer relationship management.

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Introduction

Customer churn is increasingly recognized as one of the most significant issues in the modern banking sector. In a highly digital and competitive market, customers are more informed and empowered than ever before, and thus the opportunity for customers to switch service providers if their expectations are not met is higher than ever. Although acquiring a new customer comes at a high expense, retaining existing customers is vital for long-term profitability (Datrics AI, 2023; Salesforce, n.d.).

Among the banking customer base, High Net Worth (HNW) individuals represent both a challenge and an opportunity. These banking customers often enjoy the capacity for large account balances, and larger-scale premium financial services along with personalized engagement and services. Losing these customers often results in larger financial ramifications beyond the immediate monetary loss (Michael, 2024; Singh & Anik, 2023). Non-HNW or Regular customers represent the majority of the banking portfolio in terms of number of customers. Individually, they may contribute less, but when bulk of ordinary customers churn away, the overall impact can be quite significant in eroding market share and potentially impacting operational sustainability (Brito, 2024).

Despite of increasing awareness, existing churn prediction models are at a disadvantage. Many churn prediction models merely use retrospective, static historical data which gives limited insight into certain behaviour changes. Typically, these churn prediction models lack segmentation, and they do not differentiate behaviours with segmentation which is important when considering HNW customers and ordinary customers (Peng & Peng, 2023). Machine learning (ML) methods have been gaining preferences in exploring churn prediction, however, many of the ML models have limited transparency and their decisions require more explanation (Papierski, 2023; Tran, 2023).

This study approaches customer churn as a segmented phenomenon and acknowledges that behavioural patterns, preferences, and risk profiles are distinctly different for HNW customers and Regular customers. This study focuses on using segment-based models for churn prediction and develops the adaptive churn risk scoring system to make churn models a practical, interpretable, and actionable solution to be actioned in real-time banking practice. (Malit, 2018).

The study's principal research objectives is to focus on proposing the most significant churn predictors for every customer segment, and comparing generalized versus segment-based predictions in regard to model performance, and determining a proposed cost-sensitivity learning scoring system coupled with a probability-based modified churn risk scoring system to promote the practicality of modelling decisions for HNW and Regular customers. These assumptions form the unique hypotheses for study which determines the modelling and evaluation strategy. By achieving the study objectives proposed, this paper puts forth a new contribution to the customer retention and banking analytics.

The flow of the paper after the introduction involves the literature review that provides an in-depth summary of literature around churn prediction modelling and customer segmentation. Then the methodology section highlights where data is obtained, as well as preprocessing,

modelling, evaluation and other approaches taken in the analysis. The results section discusses model performance and insights. Finally, the study is concluded with a discussion, including recommendations, potential limitations of our research, and directions for future research.

Literature Review

Customer churn is a significant problem in banking, as it influences revenue, retention strategies, and profitability over time. As Singh and Anik (2023) point out, it is more expensive to acquire a new customer than it is to keep an existing one, therefore, the prediction of churn is a strategic priority.

Baker et al (2021), have found traditional management of customer churn in banking has primarily involved either manually monitoring customer data, or rule-based systems that are fixed and reactive, which do not scale well (Michael, 2024). These approaches tend to predict churn only after the action has occurred, which does not allow for intervention (Tran, 2023).

In this context, the deployment of machine learning (ML) has emerged as a powerful mechanism to predict churn in a proactive manner. Whereas traditional approaches to customer churn would often take a sub-sample of behavioural data, ML models would be able to curate and utilise as much of the behavioural data available. From there ML can distinguish patterns, and eventually predict churn accurately. Malit (2018) and Singh and Anik, (2023), discuss the improvement of ensemble types of models like random forests and gradient boosting, over traditional models. However, these models typically lack an element of interpretability, a critical element in any field working in regulated environments such as banking, where decision making requires transparency, as well as, providing an understanding of how and why the decision was made for accountability (Peng & Peng, 2023).

Despite the improvements in modelling churn in a quantitative way, logistic regression continues to be popular in banking because it is simple and transparent (Tran, 2023). And it can demonstrate the coefficient of the model and provides information from a probability perspective and proposition for communicating with stakeholders and making real-time decisions for the relationship manager, a characteristic when using ML algorithms is not typical.

A significant gap in the research is symmetrically exploring modelling churn by segments. While several studies explore churn modelling, most have applied the model in a generalized way to accounts not accounting for the behavioural and decision-making differences between accounts and/or customers. Several studies in the recent literature (Papierski, 2023; Brito, 2024) for example, apply the context of segments based on income or engagement of customers, indicating for example, income predicts churn and that segment aware modelling based on strong engagement could model churn effectively. Despite this, few if any studies have attempted to explicitly model churn separately for High Net Worth (HNW) and Regular customers.

For instance, HNW customers in our study are defined as attributes such as an annual income of \$100,000 and above as it is one of the common ways of defining and identifying HNW accounts (Lemos, Silva, & Tabak, 2022). This identification of HNW customers based on

income is common in retail banking in crafting and deploying larger relationship roles to customers where financial value is assessed as a practical measure. This segment represents the highest potential for the bank, therefore, is of interest to model predictive churn modelling.

This literature review suggests a need for churn models that are accurate as well as interpretable, segment-aware, and operationally actionable, gaps that this research proposes to address.

Research Gap

While machine learning-based churn prediction has substantially advanced in the last decade, it is largely based on a generalized modeling approach with the vast majority of existing literature treating all banking customers as a homogeneous group, primarily ignoring the important behavioral and financial differences for engagement between customer segments, specifically, between High Net Worth (HNW) customers and regular customers. HNW customers are likely to engage in slightly different ways with their banking relationship, have higher expectations of the service level of engagement they receive from banks, and make greater revenue contributions, and all three behavioral dimensions warrant a specific churn prediction approach.

Additionally, there has been acknowledgment that while ensemble models (e.g. Random Forest, Gradient Boosting) have been shown to be better in predicting performance, they have also been known to be criticized as less interpretable. In a regulated industry like banking, this raises the concern of being unable to substantiate/model the churn reasons to decision-makers or the reason for the prediction model to act. This study addresses this matter by providing an interpretability/ feature importance framework and or framework that is segment specific to allow for practical use.

Also noteworthy, most studies related to churn prediction are mainly measured reported using accuracy metrics only without business aligned thinking incorporated, specifically around cost-sensitive learning. There are very few models presented which considers the financial impact of classifying clients incorrectly, in particular high value clients, nor deployable tools that implement the model to integrate predictive insights into strategies that relationship managers can use.

Problem Statement

This study specifically addresses the absence of, segment specific, interpretable, business-aligned machine learning framework in order to provide a churn prediction model specifically based- ten separate for (HNW) and regular customers, that integrates cost-sensitive learning (CS) and deploy a churn risk scoring approach (CRS) to assist relationship managers in proactively managing churn risk (and subsequently retention appropriateness) in order to meet organizational focus on retention, manage real-time customers-level engagement, and mitigate unnecessary relationship declines.

Methodology

Overview of Research Design

This research paper utilizes a segment-specific, business-oriented methodology in forecasting customer churn in the banking sector. The argument is based on High Net Worth (HNW) and Regular customers possessing different wealth management and financial behavior, which creates separate churn risks. Modeling these customer segments separately improves relevance and accuracy. First, the study examines baseline models of four popular machine learning algorithms namely Logistic Regression, Decision Tree, Random Forest and Gradient Boosting. To establish baseline performance, each model was trained and evaluated independently within each customer segment using a suite of exploratory classification metrics namely accuracy, precision, recall and F1 score.

After evaluating baseline model performance, Random Forest was found to be the best performing model for HNW customers, and Gradient Boosting was the best performing for Regular customers. Both models underwent additional work utilizing a cost-sensitive learning approach to address class imbalance and improve the detection rate of the minority churn class. After building the final cost-sensitive model, the study conducted a segment-specific feature importance analysis for transparency in the predictors of churn. The study also produced a churn risk scoring system that classifies customers as High, Medium or Low churn risk. The ultimate goal was to design a system that would be commercially usable. (Figure 1)

Customer Churn Prediction Overview

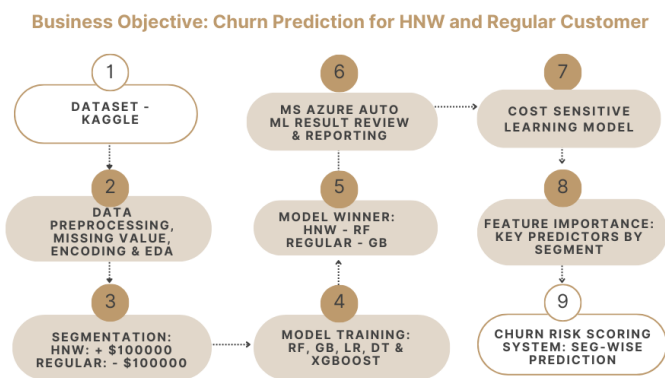


Figure 1: Research steps used in investigation of customer churn prediction

Data Collection and Preprocessing

The dataset utilized for this study included ten thousand anonymized customer records from a retail banking context. Each customer record constitutes an individual, with various features for demographic, financial, and emotional/behavioral aspects derived from simulations in the banking sector. Features included but are not limited to gender, age, geographical location, credit score, account balance, number of products held, tenure with the bank, credit card status of ownership (active), and estimated annual salary. The target variable labeled churn (where 1 = churned, 0 = no churn) attributes how the customer behaved in accordance with the classification task of this study.

Database verification of the data corpus to be used for machine learning finished with normal data quality reports. Initial exploratory analysis revealed minimum missing data, which avoided the need for major imputation applications to fill missing cells. Data preprocessing included the transformation of categorical features into number form from variables such as gender and geography into one-hot encoding suitable for any sophisticated mathematical algorithms. Continuous variables, such as credit score, age, balance, estimated salary, were standardized using a z-score normalization technique that allowed for improved convergence and comparable scales.

A major step in preparing the dataset involved partitioning the customers into two types, where the classifications of High Net Worth (HNW) or Regular customers were separated. Participants with an estimated annual salary of \$100,000 or higher would be classified as High Net Worth (HNW), which are closer outcome categories used to segment customers during the practical completion stages in retail banking. This was chosen to support the questions asking about segmented behaviour relationships if different customer segments produce divergent behavioural aspects towards churn predictors and the performance of models. This segmentation ordered the questions of the hypotheses with regard to segmented performance regarding behaviour and segmentation modelling.

Considering churn events were not that frequent in the dataset, around 20% of customers had churned, we considered the class imbalance problem while training the model through cost-sensitive learning. Modifications were made to the class weights, so that misclassifying churned customers would incur a higher penalty to the model, especially in the HNW segment since each customer has a higher total financial risk. This corresponds with the third hypothesis which states that cost-sensitive learning improves overall model performance in imbalanced class classification problems.

To provide the reader a summary of relevant features of the dataset, the following summary Table 1, provides a general description of each variable, description, and type of data prior to preprocessing.

Table 1. Summary of Features Used in the Dataset

Feature Name	Description	Data Type
Gender	Gender of the customer	Categorical

Geography	Country of residence	Categorical
Age	Age of the customer in years	Numerical
CreditScore	Customer's credit score	Numerical
Balance	Account balance in USD	Numerical
EstimatedSalary	Estimated annual salary in USD	Numerical
NumOfProducts	Number of bank products used	Numerical
HasCrCard	Whether the customer owns a credit card (Yes/No)	Binary
IsActiveMember	Whether the customer is an active member	Binary
Tenure	Number of years the customer has been with bank	Numerical
Churned	Target variable: 1 = Churned, 0 = Retained	Binary

The above table enhances transparency in the feature engineering process while facilitating ease of understanding data's structural characteristics. Beyond the information gleaned from Table 1, a churn distribution table is provided (Table 2) as a visual representation and elucidation of the class imbalance, offering additional support to the rationale for a cost-sensitive learning strategy.

Table 2. Distribution of Churned vs. Non-Churned Customers

Churn Label	Number of Customers
Retained (0)	8000
Churned (1)	2000

Exploratory Data Analysis (EDA)

The data was explored in order to comprehend how churn is distributed throughout the dataset, alongside evaluating potential behavioral and demographic features that could affect model performance. The overall churn rate was approximately 20%, which confirms class imbalance and justifies cost-sensitive modeling and evaluation strategies.

When separated through customer segment, High Net Worth (HNW) customers had a slightly higher churn rate (~21%), compared to Regular customers (~19.5%). This indicates that high-income clients are strategic and important, however, they do not exhibit more loyalty compared to Regular customers. Gender analysis exhibited that female customers were slightly more likely to churn compared to male consumers, which may indicate a different service experience or engagement strategy.

Tenure was identified as another important factor. Consumers with less than three years of a relationship with the bank were more likely to churn, which supports the assumption that new customers are more likely to be volatile and, potentially, require greater retention strategies. A similar outcome was witnessed when analyzing products owned, as single product customers were more or significantly likely to churn, compared to multiple product customers, which suggests that engagement with products is related to retention.

These preliminary observations informed what the stratified model features should be and feature weights, ensuring that the machine learning models replicates the real-world patterns of customers. All main observations are displayed in Table 3.

Table 3. Key Observations from Exploratory Data Analysis

Feature or Segment	Key Observation
Overall Churn Rate	Approximately 20% of the total customers exited the bank.
HNW vs Regular Customers	HNW customers showed a slightly higher churn rate (~21%) compared to Regular customers (~19.5%).
Gender	Female customers exhibited marginally higher churn rates than male customers.
Tenure	Customers with tenure less than 3 years were more likely to churn than long-term customers.
Number of Products	Customers holding only one product were significantly more likely to churn.

Baseline Model Development

To demonstrate the performance baseline for churn prediction, we tested four popular supervised machine-learning algorithms namely Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. The models were trained separately for each customer type, High Net Worth (HNW) and Regular, to address the unique behaviors and financial differences between the segments.

The model was developed using a stratified 80/20 train-test split, while also maintaining the class distribution of the target variable (churned). This was essential, considering there is a natural class imbalance in the data, with only ~20% of customers in the dataset having churned. All models were trained on the preprocessed features, which included standardized numeric variables and one-hot encoded categorical variables. Each model was evaluated using the following performance metrics which are accuracy, recall, precision and F1 Score.

The initial results showed the Random Forest model performed the best in the HNW model, whereas Gradient Boosting outperformed the other models in the Regular segment based on the F1 Score and recall for the churn class. These findings helped narrow down which models to enhance using cost-sensitive learning in the next phase of the study. A more detailed presentation of the model's performance metrics in the Results section (Tables 4A and 4B).

Cost Sensitive Learning Implementation

In light of the class imbalance in the datasets, wherein only approximately 20% of customers had churned, the baseline models exhibited a preference for the majority class, while lackluster performance in identifying churners. This is a barrier for business operations, especially when high-value customers are misclassified as not churners, presenting missed opportunities for intervention. To address both statistical imbalance and misclassification disparities, the models were modified to account for costs related to misclassification for those in the churned category.

The cost-sensitive learning approach during model training was used for classification at a higher misclassification cost to the churned class. This was operationalized in terms of class weights as {0: 1, 1: 3}, and through this method, misclassification of churned customers occurred three times more than misclassification of retained customers. The class weights were applied to the training of each of the two best performing models through the model identification process in the baseline evaluation, with Random Forest modeling applied to high net worth (HNW) customers and Gradient Boosting modeling applied to regular customers. Once those models were learned through accounting for more weight misclassification for the minority class, the model was learning to identify churn indications with more sensitivity, particularly in regard to financial related action (i.e., retention targets other than high net worth customers may be less valuable).

This component of the modified models played a key role to situate the machine learning and its outputs into the real-world context and its retention priorities. The results of the modified or cost-sensitive models would be presented in the section Results, as noted in the corresponding sections for recall for churned (1) and either F1 score for churned customers (1) or retained

customers (0) will highlight both increased recall for predicted churned (1) and improvement in F1 scores for churned (1) and retained (0) customers, demonstrating that the modified model reflects not (1) for a technical accuracy perspective but value (1) from an operational perspective.

Feature Importance Analysis

To improve model explainability and better connect predictive insights back to business decision-making, feature importance analysis was performed on the final cost-sensitive models for each of the customer segments. This step was particularly valuable in terms of identifying which features drove churn predictions the most and gaining assurance that the models were learning from features that actually made sense in the banking context.

For the high-net-worth (HNW) segment, feature importance was derived from a cost-sensitive random forest model. For the regular segment, it was calculated from the cost-sensitive gradient-boosting model.

The analysis indicated distinct differences in churn predictors between the segments. In the case of the HNW group, the strongest predictors were age, estimated salary, credit score, and account balance, which are rather reasonable features for this cohort as high-value clients are often looking for a tailored financial experience. Meanwhile, with respect to regular customers, age, products held in the HNW portfolio, and account balance were the strongest predictors, suggesting that depth of engagement and demographic factors are somewhat more vital to retention behaviour than for the HNW group.

Visualizations of feature importance are presented in the Results section Of the current manuscript (Figures 2 and 3).

Churn Risk Scoring System

To facilitate the operational applicability of the machine learning models, a churn risk scoring framework was consequently established. This mechanism affords utility by converting predicted churn probabilities into practical value for relationship managers responsible for customer retention.

To achieve this, churn probabilities for each customer were estimated using the probability scores generated by the `predict_proba()` function from the models. As previously stated, these churn probabilities were calculated separately for each segment using the final cost-sensitive models. In the case of the High Net Worth (HNW) segment, churn probabilities were determined using the cost-sensitive Random Forest model, while churn probabilities were obtained from the cost-sensitive Gradient Boosting model for Regular customers.

All customers have been assigned a churn risk score based on their probability score generated by the model. Risk scores for each customer were categorized into three intuitive risk tiers: High Risk (score ≥ 0.70), Medium Risk (score between 0.40 and .69) and Low Risk (score < 0.40).

This thresholding facilitates quick understanding of the churn likelihood and facilitates accordingly action by banking professionals. A risk report was constructed combining key demographic and financial features of customers (such as customer ID, age, credit score, tenure, balance, and estimated salary), with the churn risk score and corresponding risk tier.

This output can also be incorporated directly into customer relationship management (CRM) platforms, allowing for immediate prioritization of which customers to reach out to for retention purposes. This scoring system ensures that the model outputs are predictive, while simultaneously interpretable and manageable within the banking workflow. Outputs from this scoring system will be shown in the Results section (Figures 4 and 5) and demonstrates the ability to surface high-risk customers to act on in a timely manner.

Results

Baseline Model Performance by Segment

The four algorithms Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting were trained and evaluated to assess their performance in predicting customer churn. For the High Net Worth (HNW) and Regular customer segments, each model was trained and evaluated separately. Performance metrics included accuracy, precision, recall, and F1 score using hold-out test sets to assess generalizability.

The Random Forest model outperformed the other models in the HNW segment, while the Gradient Boosting model selected for the Regular segment was determined to be the superior performing model. The selections were based mainly on the F1 score and recall, as these metrics are the most relevant for imbalanced classification problems. Performance metrics for each model can be found in Table 4A (HNW) and Table 4B (Regular).

Table 4A. Model Performance – High Net Worth Customers

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.807	0.500	0.150	0.231
Decision Tree	0.791	0.459	0.466	0.463
Random Forest	0.857	0.750	0.389	0.512
Gradient Boosting	0.852	0.727	0.373	0.493

Table 4B. Model Performance – Regular Customers

Model	Accuracy	Precision	Recall	F1 Score
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Logistic Regression	0.818	0.661	0.203	0.311
Decision Tree	0.791	0.482	0.461	0.471
Random Forest	0.846	0.731	0.376	0.497
Gradient Boosting	0.848	0.731	0.391	0.510

Cost-Sensitive Model Performance

To address a challenge regarding sensitivity to churned customers, cost-sensitive learning was applied to the best model of each segment. The class weights (i.e., `'class_weight={0: 1, 1: 3}'`) were modified to assign a weight of three to churned customers, relative to a retained customer prediction. This modification assisted the model in predicting the minority class correctly and more emphasis on the churn predictions.

The cost-sensitive Gradient Boosting model for Regular demonstrated strong improvement in recall and F1 measurement in particular for the churn class but for the cost-sensitive Random Forest model for HNW did not show an improvement, on the other hand, there was a decrease in all parameters. Performance measurements of these cost-based models are provided in Table 5.

Table 5. Cost-Sensitive Model Performance by Segment

Segment	Model	Precision (Churn Class)	Recall (Churn Class)	F1 Score (Churn Class)	Accuracy
High Net Worth	Random Forest	0.71	0.34	0.46	0.85
Regular	Gradient Boosting	0.49	0.69	0.57	0.79

Evaluation of Feature Importance

In the interest of increasing transparency, and in the spirit of business relevance, feature importance was evaluated for both cost-sensitive models. The Random Forest model identified age, estimated salary, credit score, and balance as the highest predictor importance values in the HNW segment. The Gradient Boosting model identified age, number of products, and balance as the most relevant predictors of churn for the Regular customer segment, as seen in Figures 2 and 3, which respectively show the ranked feature importance maps for each customer segment. This exercise affirms the decision to model the segments separately, while lending credibility to business relevance of the predictors employed to inform price change segmentation.

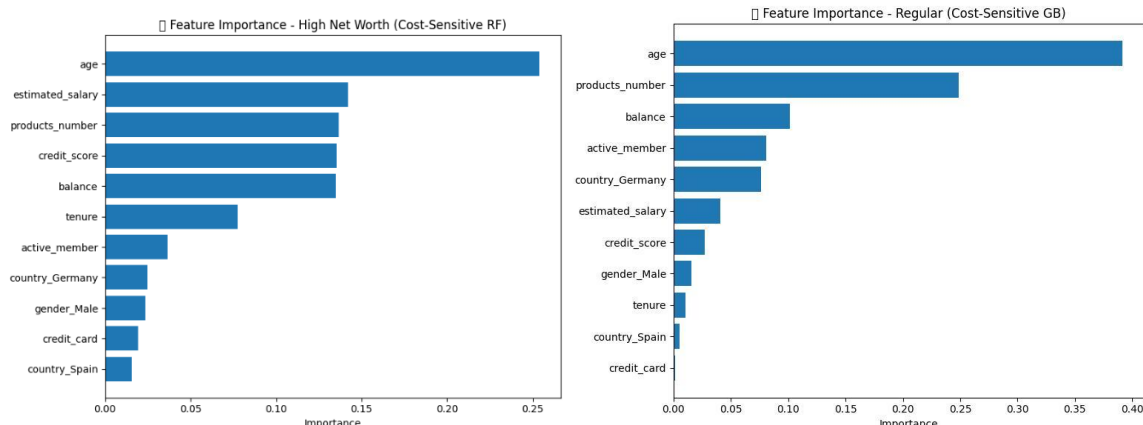


Figure 2 (Left): Feature Importance – High Net Worth Customers (Random Forest) and Figure 3 (Right): Feature Importance – Regular Customers (Gradient Boost)

Churn Risk Scoring System

In order to translate model predictions into operational insights, a churn risk scoring system was developed. The final cost-sensitive models were used to create probability scores that indicate the likelihood that each customer will churn. The probability scores were then used to prioritize customers and group them into risk segments – High (≥ 0.70), Medium (0.40–0.69), and Low (< 0.40).

Figure 4 illustrates an output from the scoring system designed for the HNW segment and identified the top 15 high-risk customers alongside meaningful demographic and financial information. Figure 5 offers a visual representation of the distribution of churn risk scores across each Tiers.

HNW Churn Risk Report (Top 15 Customers):

	customer_id	age	gender	balance	tenure	credit_score	estimated_salary	churn_risk_score	risk_level
1608	15764604	35	Female	164769.02	7	586	119814.25	0.94	High
1228	15704442	53	Female	169406.33	9	672	147311.47	0.93	High
3468	15665283	57	Female	72092.95	7	610	113228.82	0.90	High
3281	15571869	50	Female	112650.89	4	669	166386.22	0.90	High
3632	15623210	55	Female	149349.58	8	484	137519.92	0.90	High
4665	15623989	54	Male	0.00	3	435	156910.46	0.90	High
984	15787189	60	Male	134250.17	8	824	153046.16	0.90	High
287	15775318	51	Female	154962.99	3	590	191932.27	0.89	High
1287	15592736	54	Male	102994.04	5	551	176680.16	0.88	High
782	15649121	52	Male	0.00	3	665	116137.01	0.88	High
3573	15600975	54	Female	150005.38	4	556	157015.50	0.87	High
4183	15677336	57	Male	120043.13	1	557	132370.75	0.87	High
1927	15602797	49	Female	110132.55	5	645	187689.91	0.87	High
3087	15720371	51	Female	0.00	3	652	173989.47	0.87	High
73	15713483	52	Male	0.00	2	793	159123.82	0.87	High

Figure 4: HNW Churn risk report (Top 15 customers)

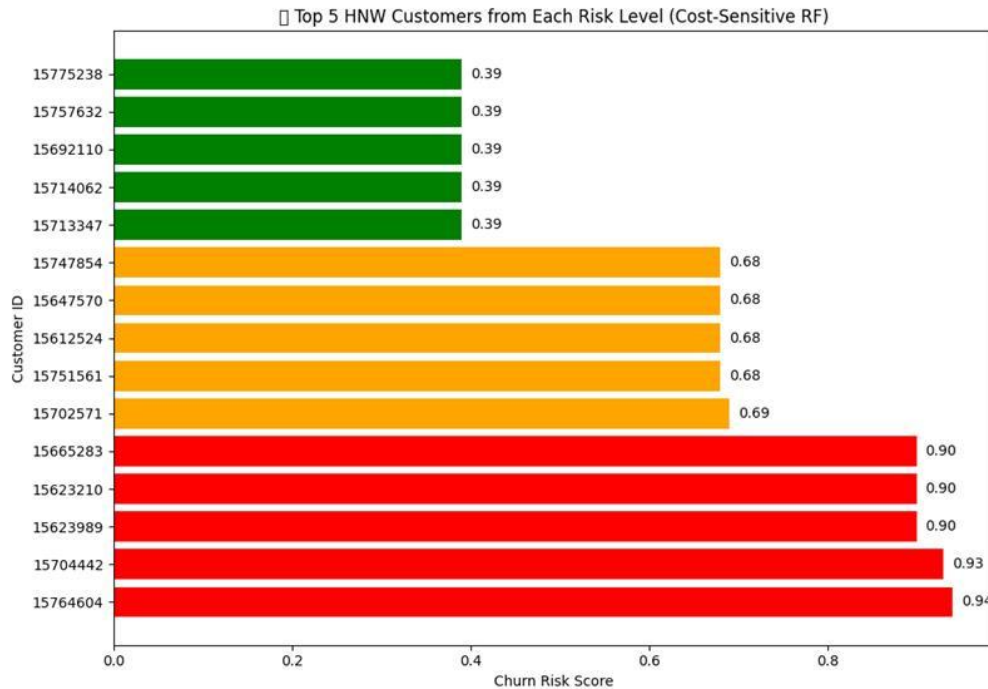


Figure 5 : Graphical Representation of top 5 HNW customers

The scoring model provides relationship managers with the ability to proactively target high-risk customers with tailored interventions and prioritize outreach based on risk-band.

Discussion

This paper sought to determine if customer churn in a banking context could be better predicted through segment-specific, interpretable, and business-facing machine learning models. Findings suggest a positive direction, notwithstanding a few unanticipated results that may merit further investigation.

A noteworthy outcome is that the non-cost-sensitive Random Forest model for High Net Worth (HNW) customers returned better F1 score versus the cost-sensitive model. This is not a surprising finding given certain trade-offs are very common in imbalanced classification problems. Here, by increasing the recall of a minority class (the churned customers), the precision of the model decreased, reducing the F1 score. While this was the case, the cost-sensitive model did improve the ability of the model to identify churners. This qualifies as perhaps even better, from a business perspective, for higher-value clients where the loss associated with not acting on the churn is much higher. For this reason, the cost-sensitive version was the version retained as the final model, for alignment with real-world retention objectives.

Also, the Regular customer segment benefited from cost-sensitive learning in both recall and F1 score parameters for the customer segment. This also adds to the literature demonstrating that tuning model sensitivity is valuable in the face of class imbalance. Furthermore, this segment also provided churn attribute predictors that were quite different to the HNW segment, in terms of number of products held and tenure, which again reinforced the decision to pursue segment-specific modeling strategies.

Another important factor in this research study is the potential for translating predictions into predictive practice. The churn risk scoring tool demonstrated what probability-based model outputs could look like when articulated as actionable customer profiles, defined by risk tiers. This articulation offers transparency and usability from the perspective of relationship managers, operating in reality, that can be tie into processes to focus retention activity on higher risk customers. Importantly for strategic decision making, some key demographic and financial risk features were presented in the scoring report.

There are some limitations of the study to recognize. Firstly, predictive performance of machine learning models may differ a great deal depending on sampling variability, hyperparameter tuning, and feature selection in the models. Further, while feature importance was utilized to increase interpretability of the model, and left the door open to return to causal relationships between features variable and churn behavior, in and of itself, it does not affirmatively demonstrate the causal relationships. Further, as the models were all trained on a single dataset rife with methodology limitations, generalizability will be about other banks or customer populations.

Nonetheless, the current research demonstrates a strong case for segmentation-aware, cost-sensitive, and deployable approaches to customer churn prediction frameworks that continue to balance predictive performance into the true (real-world) utility. The combination of having business segmentation relevance, interpretable models, and more operational scoring tools and reports, move the study along the research continuum from pure technical analysis toward the application and engagement of actionable customer read on, strategic response navigation.

Conclusion

This research investigated a segment-specific, cost-sensitive, and deployable machine learning approach for predicting customer churn in the banking industry. By separately modeling High Net Worth (HNW) and Regular customers, this research accounted for the differences in behavior and finance that machine learning approaches sometimes ignore. It showed that customized modeling resulted in useful insights that were consistent with customer profiles and business needs.

Using cost-sensitive learning to improve the model performance was one of the highlights of this paper . This approach did incur some trade-off for the HNW group in terms of F1 score, but it increased recall in important ways that further justified its use in contexts, where the business impacts related to false negatives were material. For the Regular customer group,

cost-sensitive Gradient Boosting produced a combination of higher recall and F1 scores, which increased the relevance and widely applicable advantages of cost-sensitive first order learning. The other significant contribution of this work was the churn risk scoring system that was conceptualized, which ensured that the output from different models could be actionable and implemented. Through the scoring system, probabilities could be sorted into risk bands, and churn drivers could be visualized in a way that would enable relationship managers to proactively contact at-risk customers. The operationalization of machine learning and churn research typically does not exist beyond an academic paper, especially in more traditional fields where implementing the model is challenging.

While the models and scoring capabilities showed possible promise for practical implementation, several limitations were evident, such as a single data source reliance and potential model variance. Future studies could take advantage of the framework around including real time behavioral features, develop a causal inference approach or pilot the framework at several institutions for generalizability.

In summary, this study provides an interpretable and practical machine learning framework to predict churn for the banking context, generate insight into specific segments, prioritize the highest risk customers, and deploy retention methods all with near real time scoring. This study addressed a gap between more advanced analytics and real world impact and contributes to both academic literature and industry practice.

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